

Spiral Line-Field Models and Kernel Regression for Chiral Data

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Abstract

We compare global parametric spiral vector fields with RKHS/Gaussian-process kernel regressions for cleaned image-derived chiral line-field data. Each observation is a segment midpoint $z_i = (x_i, y_i)$ together with an axial direction $\hat{\alpha}_i \equiv \hat{\alpha}_i + \pi$. The analysis uses local axial-consistency cleaning, held-out axial residual sum of squares (RSS), optimized Gaussian-process evidence for doubled-angle embeddings, and BIC approximations for finite-dimensional parametric models. The multiplicative spatial-angular RKHS kernel is preferred by both held-out RSS and evidence-style model comparison on the 39-dataset cleaned stress test.

1 Data and Parametric Spiral Model

For each dataset, the raw perpendicular segments are converted to midpoint locations $z_i = (x_i, y_i)$ and observed line-field angles $\hat{\alpha}_i = \text{atan2}(\Delta y_i, \Delta x_i)$. Coordinates are centered and the image y -axis is inverted to obtain a Cartesian convention. Since the segments are axial, residuals identify α and $\alpha + \pi$:

$$\Delta_i^{\text{ax}}(\mathcal{M}) = \frac{1}{2} \text{atan2}\{\sin(2[\hat{\alpha}_i - \hat{\alpha}_{\mathcal{M}}(z_i)]), \cos(2[\hat{\alpha}_i - \hat{\alpha}_{\mathcal{M}}(z_i)])\}. \quad (1)$$

The cleaned dataset is defined reproducibly by a central circular region of interest (ROI): a segment is retained only if its midpoint lies inside the central disk of the corresponding 150×150 arrow map, with center (75, 75) and radius 58.5. This rule preserves the central circular line field and removes peripheral segments outside the circle. It retained a median fraction 0.992 of segments across the 39 datasets, with range 0.414–1.000 and median cleaned sample size 123.

The parametric family is derived from the scalar potential

$$V_\theta(r, \phi) = Ag_p(r) - b\phi, \quad g_p(r) = \begin{cases} \log r, & p = 0, \\ r^p/p, & p \neq 0, \end{cases} \quad (2)$$

with polar coordinates (r, ϕ) . Its vector field is

$$F_\theta(r, \phi) = \nabla V_\theta(r, \phi) = Ar^{p-1}e_r - \frac{b}{r}e_\phi, \quad \rho_\theta(r) = \|F_\theta(r, \phi)\| = \left(A^2r^{2p-2} + \frac{b^2}{r^2}\right)^{1/2}. \quad (3)$$

The fixed exponent models are $p = 0$ (logarithmic), $p = 1$ (Archimedean), and $p = 2$ (Fermat). We also fit a continuous exponent model with $p \in (-1, 3)$. The angular noise motivation is the standard delta-method approximation for the angle of a noisy two-dimensional vector: if $W_i = F_{\theta_0}(z_i) + \varepsilon_i$ and $\varepsilon_i \sim N(0, \sigma^2 I_2)$, then $\text{Var}(\hat{\alpha}_i) \approx \sigma^2 / \rho_{\theta_0}(z_i)^2$ [6, 2]. Thus weak-field regions are intrinsically less informative, but model scoring is axial rather than signed.

2 Exact Kernel Models

The kernel regressions use the same observations but replace the single global spiral law by a smooth non-parametric covariance structure. Let $d(z, z') = \|z - z'\|_2$ and let $\vartheta(z) = \text{atan2}(y, x)$ be the polar angle of the location. The three kernel families used in the stress test are

$$K_{\text{RBF}}(z, z'; \ell) = \exp\left[-\frac{d(z, z')^2}{2\ell^2}\right], \quad (4)$$

$$K_{\text{unif}}(z, z'; \ell) = \mathbf{1}\{d(z, z') \leq \ell\}, \quad (5)$$

$$K_{\text{mult}}(z, z'; \ell, \kappa) = \exp\left[-\frac{d(z, z')^2}{2\ell^2}\right] \exp[\kappa \cos\{\vartheta(z) - \vartheta(z')\}]. \quad (6)$$

The Gaussian RBF kernel controls spatial smoothness through the length scale ℓ . The uniform kernel is a compact local-neighborhood smoother; when interpreted as a covariance matrix it is stabilized numerically by the nugget term described below. The multiplicative kernel augments spatial similarity with polar-angle similarity. Importantly, the angular term in (6) uses the geometry of the location, not the observed response angle, and therefore does not leak held-out outcomes into the predictor. In the implementation used for these results, the kernel amplitude is fixed to one and the observation-noise variance is swept explicitly.

The Gaussian RBF kernel is universal on compact subsets of Euclidean space: its RKHS is dense in the continuous functions under the uniform norm [5]. The angular factor in (6) is the von-Mises kernel on S^1 ; for $\kappa > 0$ its Fourier coefficients are strictly positive, so it is also universal on the circle. Hence the product kernel is universal on compact subsets of the product geometry (z, ϑ) , and its restriction to the observed spatial domain remains a flexible non-parametric prior. This universality statement does not apply to the uniform indicator kernel, which is used here as a local smoothing baseline rather than as a theoretically universal covariance.

For a kernel K_θ , the Gaussian-process covariance matrix is

$$C_\theta = K_\theta(X, X) + \sigma_n^2 I_n. \quad (7)$$

The line-field response is represented by the doubled-angle embedding $u_i = (\cos 2\hat{\alpha}_i, \sin 2\hat{\alpha}_i)$. The GP fits the two coordinates independently with the same covariance C_θ , and predictions are mapped back to an axial angle by $\hat{\alpha}(z) = \frac{1}{2} \text{atan2}\{\hat{u}_2(z), \hat{u}_1(z)\}$. The bandwidth grid is dataset-adaptive rather than fixed: if $s_X = \max\{\text{range}(x), \text{range}(y), 1\}$ and d_{nn} is the median nearest-neighbor spacing, then

$$\ell \in \text{geospace}\{\max(d_{\text{nn}}/2, s_X/80, 10^{-3}), \max(1.5s_X, 2d_{\text{nn}}, 4s_X/80), 14\}.$$

The multiplicative kernel additionally sweeps $\kappa \in \{0, 0.5, 1, 2, 4, 8, 16\}$, and each model sweeps eight dataset-adaptive values of σ_n .

3 RSS, Evidence, BIC, and Bayes Factors

The predictive metric is held-out axial RSS,

$$\text{RSS}_{\text{test}}(\mathcal{M}) = \frac{1}{K} \sum_{k=1}^K \frac{1}{|I_k|} \sum_{i \in I_k} \{\Delta_i^{\text{ax}}(\mathcal{M}_{-k})\}^2, \quad (8)$$

where $\mathcal{M}_{-k}$ is fit without the k th fold. RSS estimates predictive risk; it does not directly penalize model dimension except through held-out performance.

The evidence score for the RKHS/GP models is the optimized log marginal likelihood of the two-dimensional doubled-angle response matrix $U \in \mathbb{R}^{n \times 2}$ [3]:

$$\log p(U \mid X, \theta) = -\frac{1}{2} \text{tr}(U^\top C_\theta^{-1} U) - \log |C_\theta| - n \log(2\pi). \quad (9)$$

The first term is data fit and the log-determinant term is the Occam factor. For the finite-dimensional parametric models, we use the BIC approximation [4]

$$\text{BIC}(\mathcal{M}) = k_{\mathcal{M}} \log n - 2 \log L(\hat{\theta}_{\mathcal{M}}), \quad \log p(y \mid \mathcal{M}) \approx -\frac{1}{2} \text{BIC}(\mathcal{M}). \quad (10)$$

Bayes factors are computed from aggregate evidence over independent datasets [1]:

$$\text{BF}_{m, m_\star} = \exp\{E_m^{\text{tot}} - E_{m_\star}^{\text{tot}}\}, \quad E_m^{\text{tot}} = \sum_{j=1}^{39} E_{mj}. \quad (11)$$

Because the evidence gaps are extremely large, we report $\log_{10}$ BF rather than the underflow-prone raw Bayes factor.

4 Results

Tables 1 and 2 report the 39-dataset central-ROI axial stress test. The multiplicative RBF–von-Mises RKHS is the leading model under held-out RSS and aggregate evidence. The parametric spirals are much more competitive under evidence after removing peripheral segments, although they remain far weaker under held-out predictive RSS.

Model	Mean test RSS	Median test RSS	Mean MAE
Multiplicative RBF-VM RKHS	0.1079	0.1060	11.50°
Gaussian RBF RKHS	0.1291	0.1160	12.54°
Uniform RKHS	0.1302	0.1176	12.81°
Parametric $p = 1$ (Archimedean)	0.6783	0.6742	39.88°
Parametric $p = 2$ (Fermat)	0.6828	0.6670	40.11°
Parametric $p = 0$ (Logarithmic)	0.6889	0.6875	40.14°
Parametric $p \in (-1, 3)$	0.6953	0.6789	40.33°

Table 1: Held-out predictive comparison on cleaned data. RSS is the mean squared axial residual per held-out point.

Model	Mean evidence	Total evidence	$\log_{10}$ BF
Multiplicative RBF-VM RKHS	-137.867	-5376.807	0.000
Parametric $p = 1$ (Archimedean)	-148.523	-5792.401	-180.490
Parametric $p = 2$ (Fermat)	-148.781	-5802.467	-184.862
Parametric $p = 0$ (Logarithmic)	-148.951	-5809.108	-187.746
Parametric $p \in (-1, 3)$	-149.981	-5849.265	-205.186
Gaussian RBF RKHS	-152.740	-5956.851	-251.910
Uniform RKHS	-261.668	-10205.062	-2096.884

Table 2: Evidence-style comparison on cleaned data. Kernel rows use optimized GP marginal likelihood for the doubled-angle embedding; parametric rows use $-\text{BIC}/2$ from axial residuals. Bayes factors are relative to the multiplicative RBF-VM RKHS model.

5 Interpretation and Caveats

The agreement between held-out axial RSS and evidence indicates that the multiplicative spatial-angular kernel is not merely overfitting training data; it also predicts held-out line-field directions best. The result is plausible statistically: a single global spiral exponent cannot adapt to local deviations in the observed field, whereas the multiplicative kernel borrows strength from nearby points with similar polar position.

Two caveats remain. First, Table 2 combines optimized GP marginal likelihoods for a two-output line embedding with BIC approximations for parametric axial residuals. This is a reasonable evidence-style comparison, but it is not a fully Bayesian analysis with matched priors over all finite- and infinite-dimensional hyperparameters. Second, the coherence filter is deliberately local and non-parametric; it removes isolated orientation outliers but does not prove that the remaining data follow any single global spiral law.

References

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